

Pixel Microlensing in M31 Using Angstrom Difference Imaging

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Scientific Motivation

Aim: Identify microlensing candidates in M31 bulge and extract the lens mass using data from the Angstrom Project

- Microlensing detects mass, not light (Crotts, 1992)
 - Detect dark matter candidates
- M31 bulge is ideal for event detection (Kerins et al., 2001)
 - M31 Bulge has high stellar density → pixel lensing

Part I: Microlensing and Pixel Lensing Theory

Aim: Describe the underlying theory for gravitational microlensing and pixel lensing

Microlensing magnification

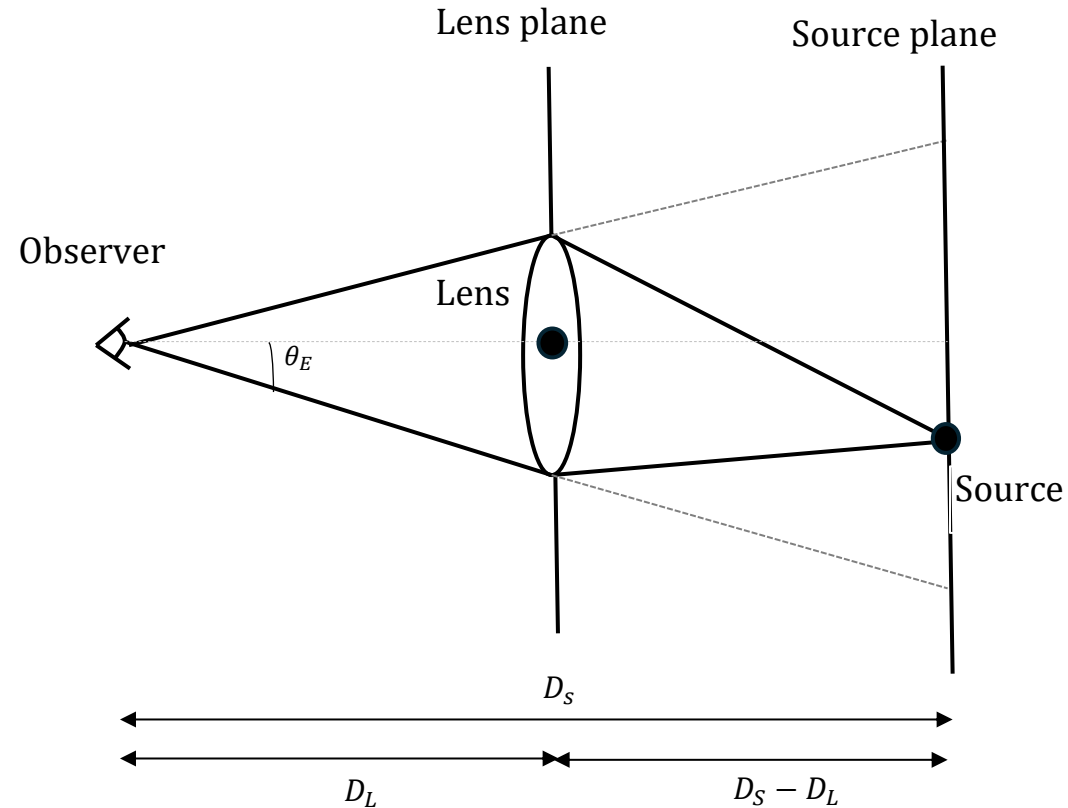
The Einstein radius is given by

$$\theta_E = \left(\frac{4GM}{c^2} \frac{D_S - D_L}{D_S D_L} \right)^{1/2}$$

Standard PSPL model for unresolved sources (Paczynski, 1986):

$$A(t) = \frac{u^2 + 2}{u\sqrt{u^2 + 4}}, \quad u(t) = \sqrt{u_0^2 + \left(\frac{t - t_0}{t_E} \right)^2}$$

- A is the magnification factor
- u is the impact parameter (ratio of the source-lens angular separation to θ_E)
- u_0 is the minimum impact parameter, occurring at time t_0
- t_E is the Einstein radius crossing time



Standard PSPL Modelling

Standard PSPL model in pixel lensing:

$$F(t) = [A(t)f_b + (1 - f_b)]F_{\text{PSF}}$$

with $A(t) \equiv A(t, t_0, u_0, t_E)$

(Paczynski, 1986)

The peak shape is governed by three parameters:

- u_0 : minimum impact parameter
 - t_E : Einstein timescale
 - f_b : blending fraction
-
- Strong correlation coefficients between f_b, t_E, u_0 (magnitudes > 0.98)

High Magnification Limit

For high magnifications of $A_0 > 10$ (Griest and Safizadeh, 1998), the PSPL model reduces to:

$$F(t) = F_{\text{base}}(t) + \Delta F_0 \left[1 + 12 \left(\frac{t - t_0}{t_{1/2}} \right)^2 \right]^{-1/2}$$

- $t_{1/2}$ alone describes the width of the peak
- Flux rise is defined relative to the baseline, removing the need for f_b

Correlation coefficients are consistently less than 0.5 across all parameter pairs

Part II: Setting up Difference Image Analysis

Aim: Produce a high-quality reference image

Difference Image Analysis

Purpose: Remove static background so only real flux changes remain

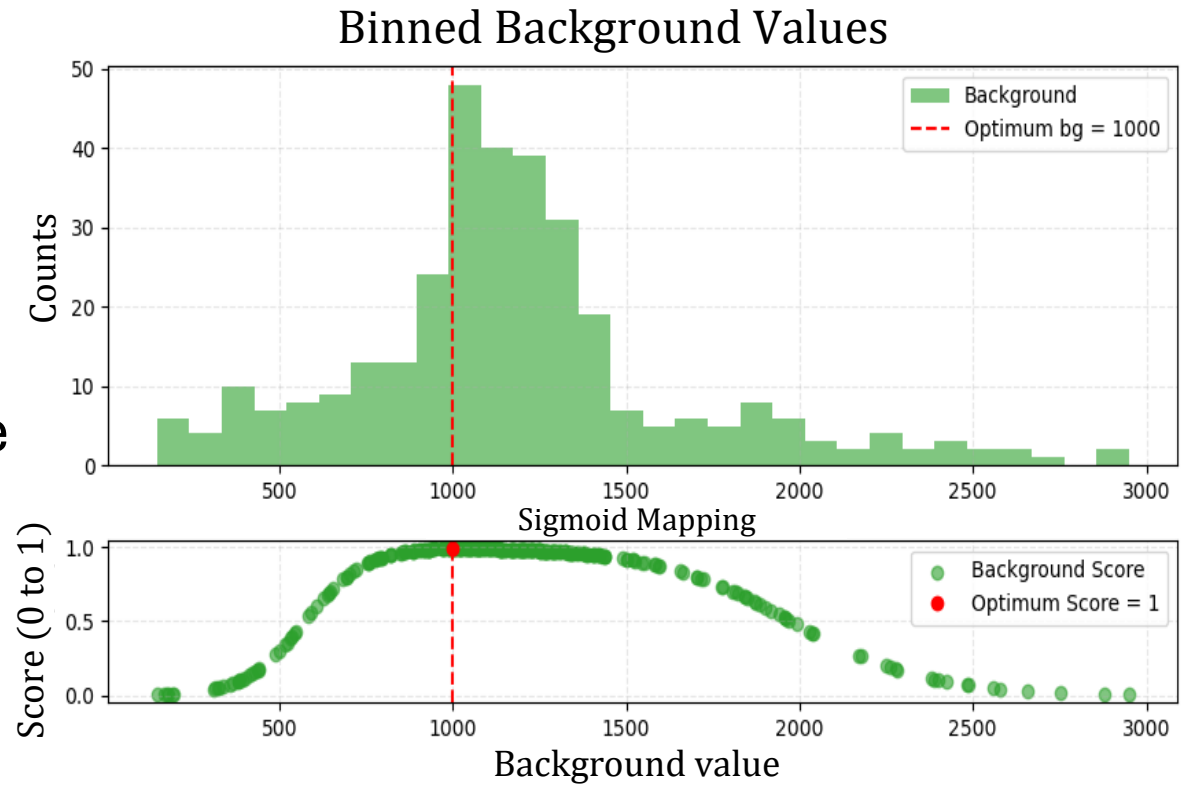
- We use a method proposed by Alard & Lupton (1998)
 - Pick a high-quality reference image, R
 - Transform R using a kernel K and background function B
 - K is formed of Gaussian components multiplied by 2D polynomials
 - B is a 2D polynomial
 - PSF, photometric scale and background match the target image T
- Difference image, D , is found by

$$\sum_i D_i^2 = \min \sum_i (R'_i - T_i + B_i)^2, \quad R' = R \otimes K$$

Reference Image Construction

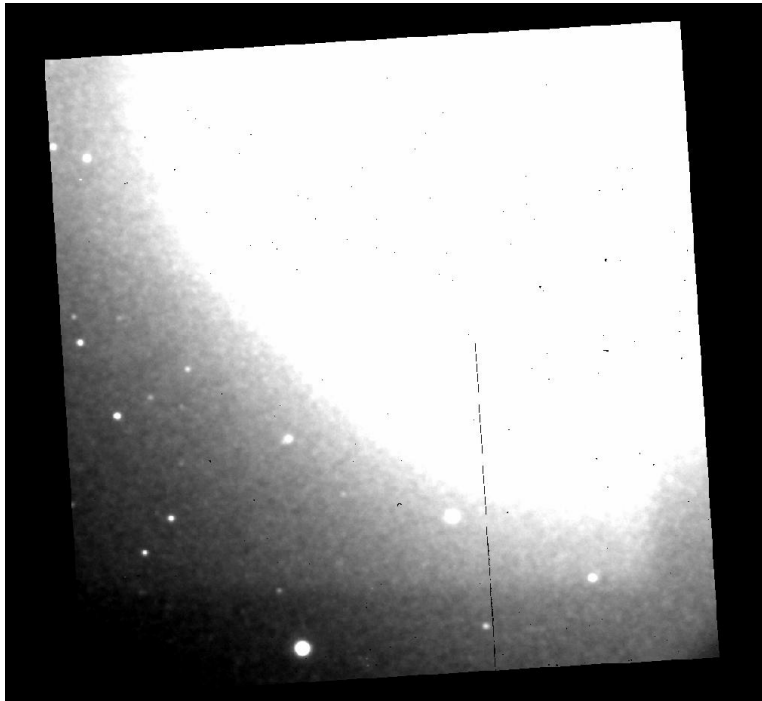
1. Give a normalised score (f) for each category per image
 - Categories: seeing, airmass, background level, background std
 - Using a sigmoid mapping from the optimum value to penalise extremes
2. Define quality factor for a given image as geometric mean of scores:

$$Q = \prod_i f_i$$

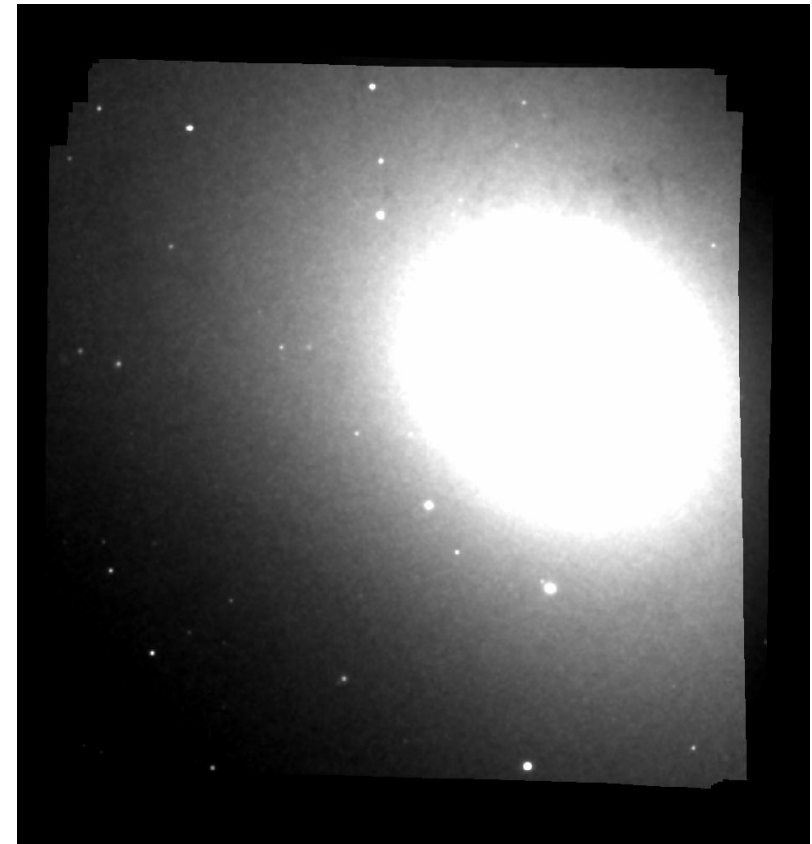


Reference Image

Manually remove defective and/or rotated images, e.g.,:



**Final reference image formed of *top*
11 images:**



Part III: Determining Background Model and Kernel Parameter Sizes

Aim: Optimise the sizes of parameters for image subtraction

DIA Parameter Size Optimisation

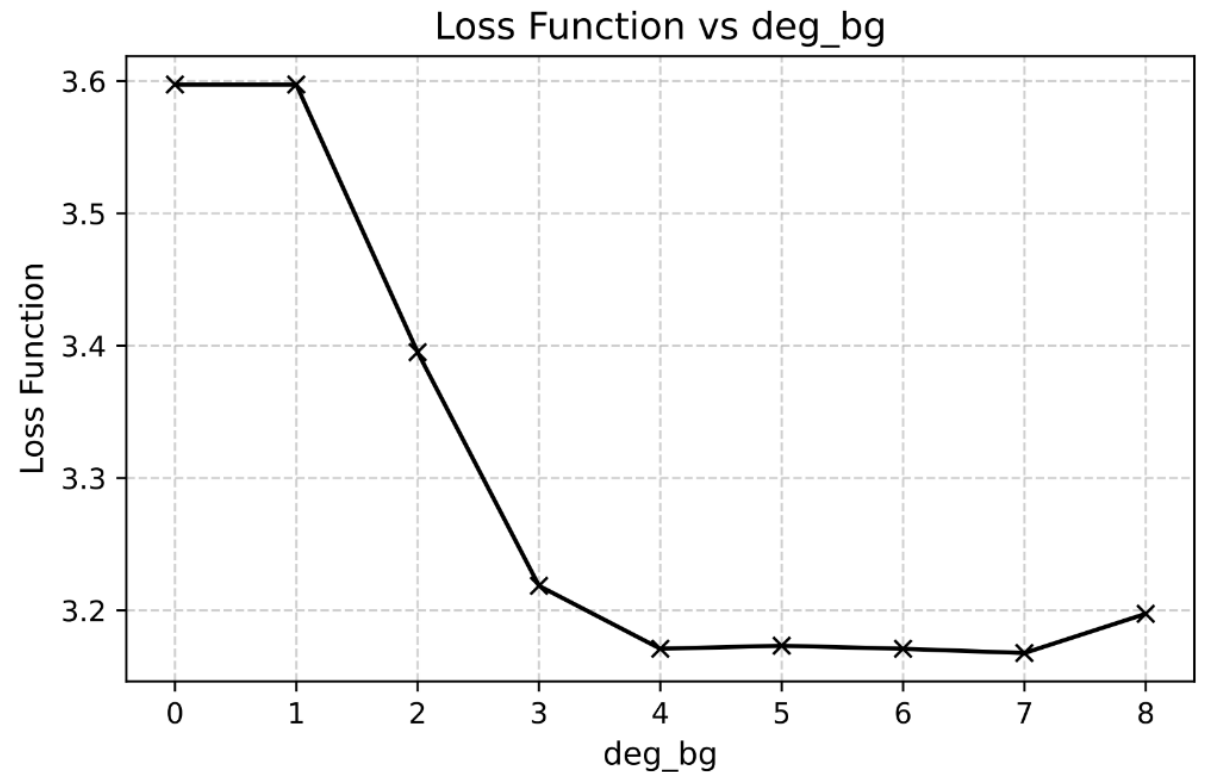
Need to determine optimal kernel parametrisation

deg_bg	n_gauss	deg_gauss	sigma_gauss
Background 2D polynomial degree	Number of gaussian basis functions for the kernel	Gaussian polynomial degrees	Gaussian widths

Started with background as this was weakly correlated with the other parameters

Background Polynomial Optimisation

- Calculate the combined z-score of mean and scatter to get loss function.
- Improvement saturates at $\text{deg_bg} = 4$.



Correlated Parameters

For each `n_gauss`, correlated parameters are optimised using a two-stage approach on the loss function:

1. Global stochastic search

- Accepts uphill movements
- Avoids local minima

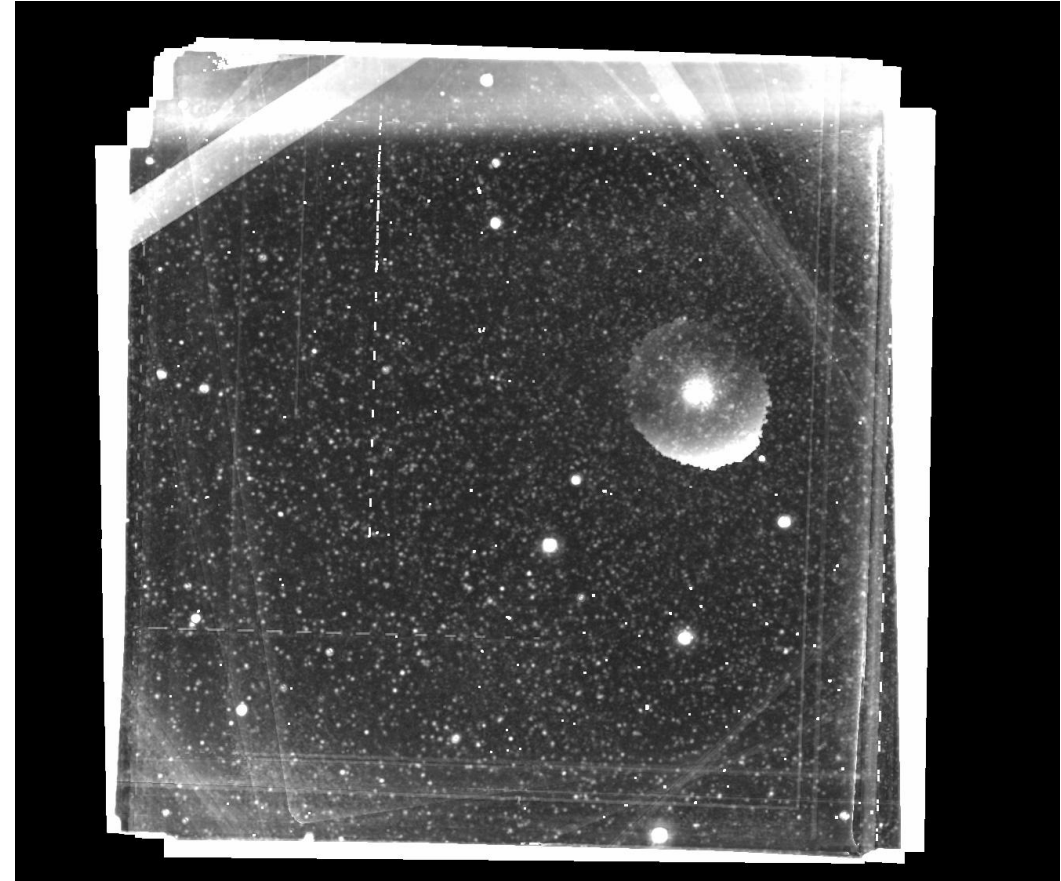
2. Local refinement – Powell's method

- Direction-set optimisation
- Finds minima precisely

Outcome: `n_gauss` = 2 (smaller parameter space = easier global convergence)

Variability Map

- Construct map by summing |difference image| for each pixel
- Apply flux threshold $\rightarrow$ suppress pixels dominated by noise
- Identify 4 candidate pixels for detailed modelling

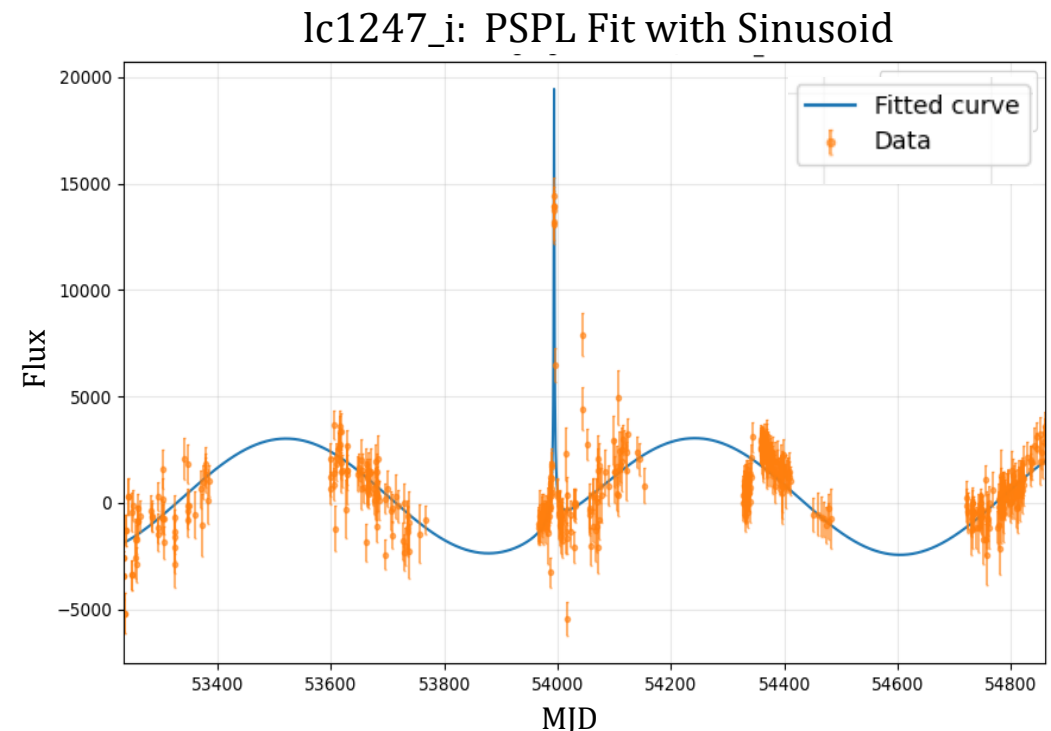
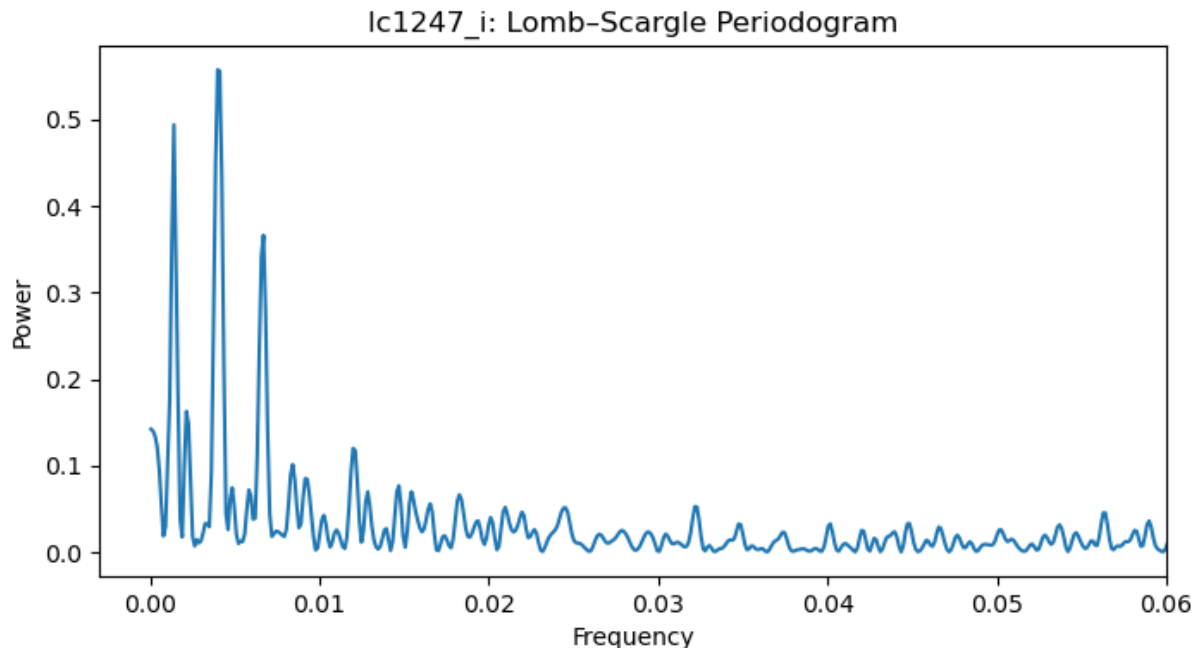


Part IVa: PSPL-like Case Studies

Aim: Analyse light curves for two PSPL microlensing-like light curves

Pixel 1: Microlensing + Periodicity

- Strong variable baseline
- Lomb-Scargle analysis (variation of Fourier transform) showed 1 sinusoidal term to baseline gave best model
- Symmetric peak, non-periodicity of peak, narrow event time range ➡ suggests it's likely to be microlensing events
- PSPL + sinusoidal fit gave $\chi_r^2 = 1.8$



Pixel 1: Microlensing + Periodicity

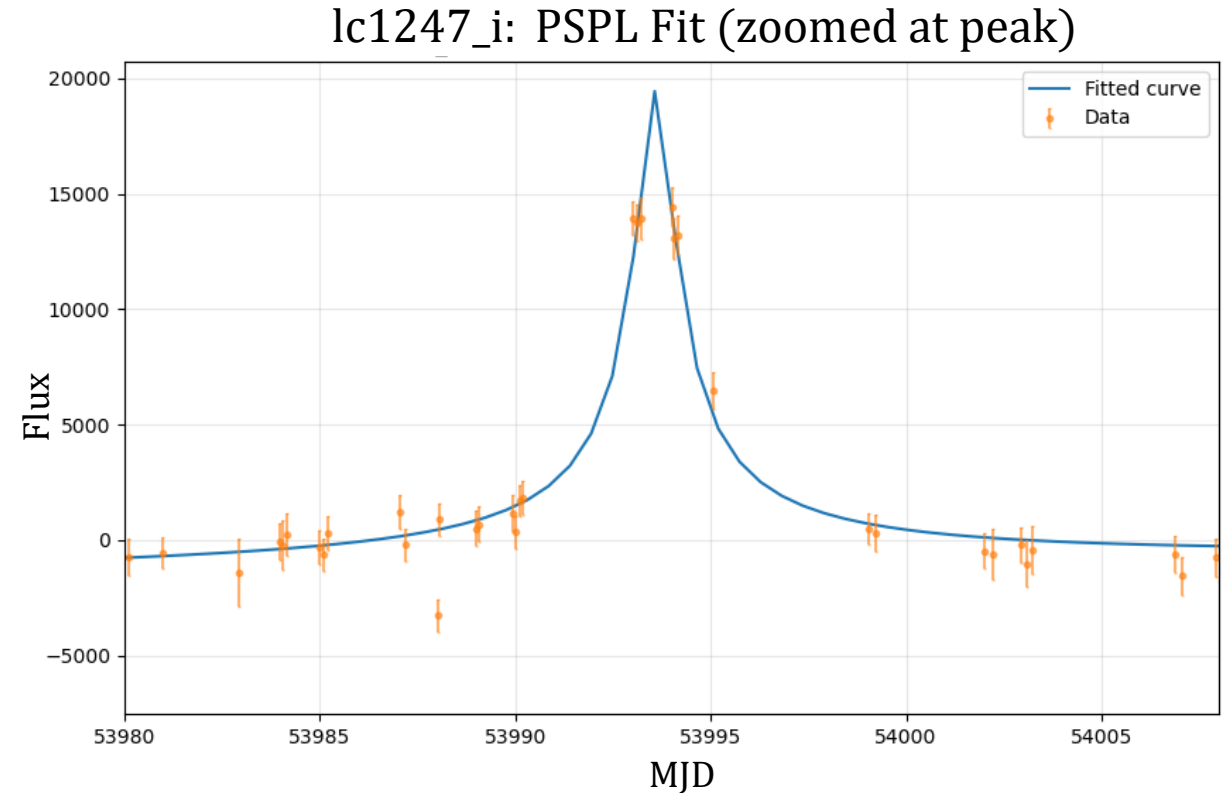
Assuming:

- Bulge self-lensing
 - Lens-source distance = (2 ± 1) kpc
- Relative velocity = (130 ± 50) km/s

$$M_{\text{lens}} = (8 \pm 6) \times 10^{-3} M_{\odot}$$

Sub-stellar object:

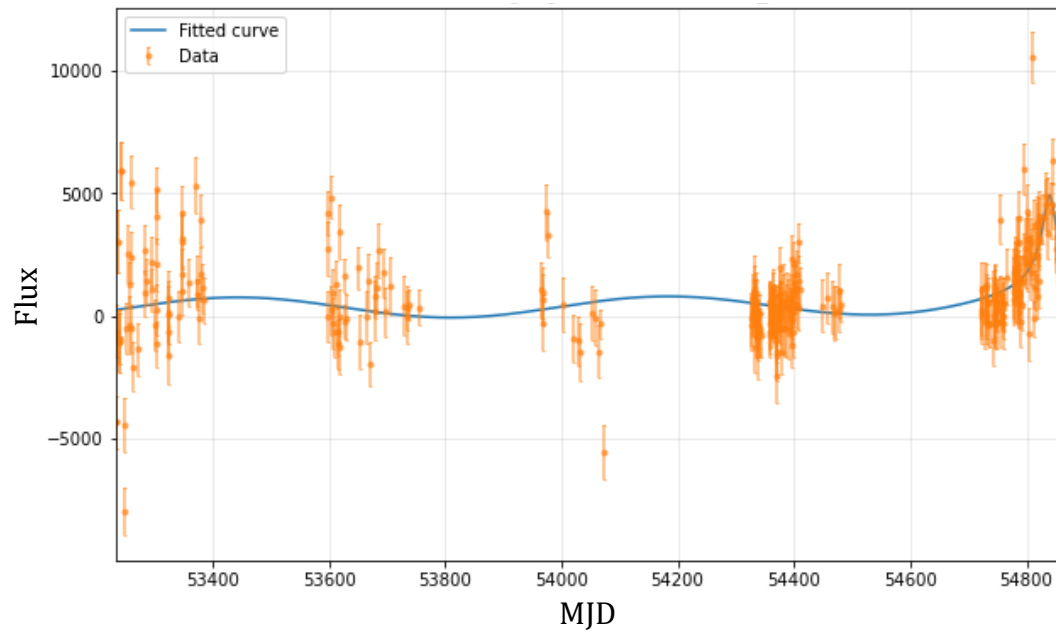
- *brown dwarf (Chabrier, 2003) or*
- *large exoplanet (Sumi et al., 2011)*



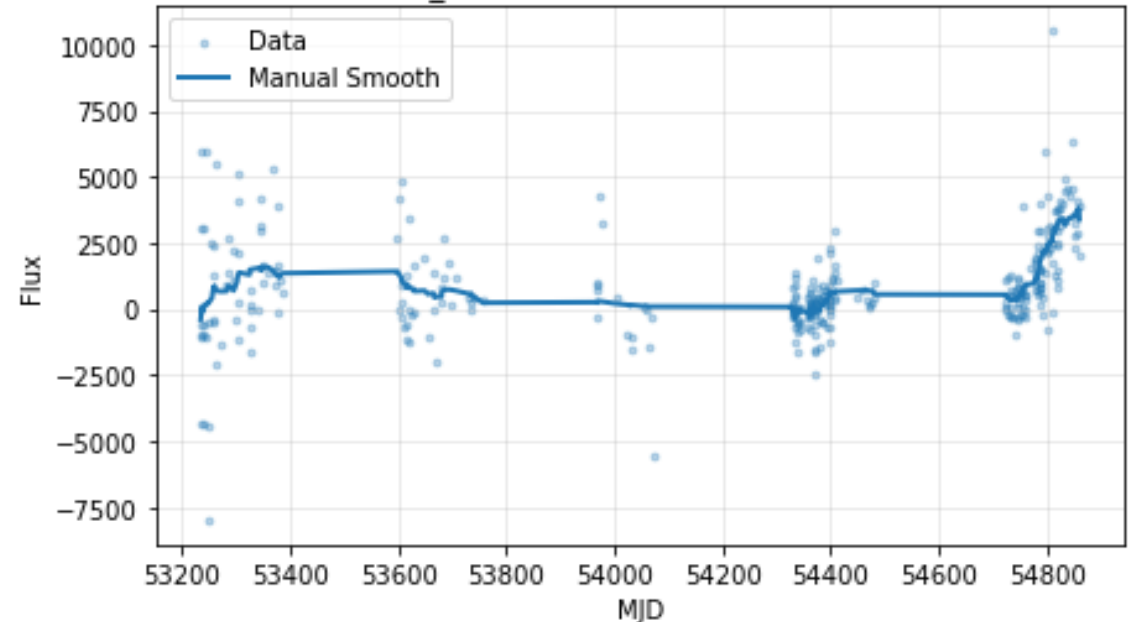
Pixel 4: Late PSPL-like Event

- Large variation in flux $\rightarrow$ global χ_r^2 (2.61) and trend unclear
 - Smoothing data reveals microlensing-like curve at end of survey
 - Symmetric peak, non-periodicity of peak. However the duration of event is wider than previous
- ➡ The mass of lens would be much bigger than pixel 1 assuming it's microlensing.

lc4762_iv: PSPL Fit with Sinusoid



lc4762_iv: Smoothed Data (window = 30)



Pixel 4: Late PSPL-like Event

Restrict to fitting to this late peak

1. Bulge-bulge microlensing

$$M_{\text{lens}} = (5 \pm 4) M_{\odot}$$

Bulge-bulge dominates (Baltz et al., 2003), but is $P(\text{bulge-bulge} | (5 \pm 4) M_{\odot})$ likely?

2. Bulge-disc lensing gives more typical lens mass

Lens-source distance = (10 ± 5) kpc and *relative velocity* = (200 ± 70) kpc:

$$M_{\text{lens}} = (2.2 \pm 1.8) M_{\odot}$$

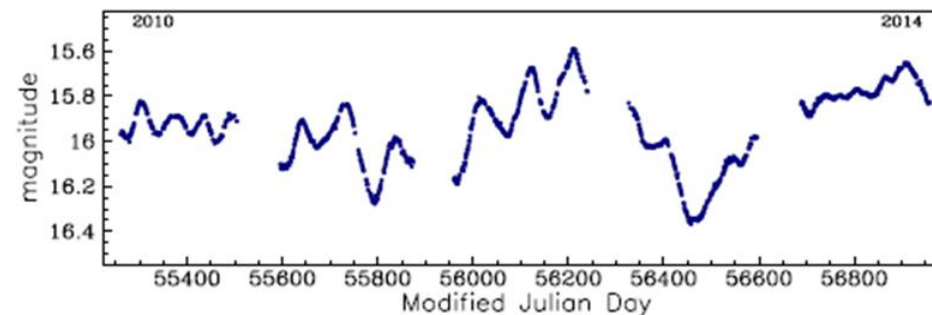
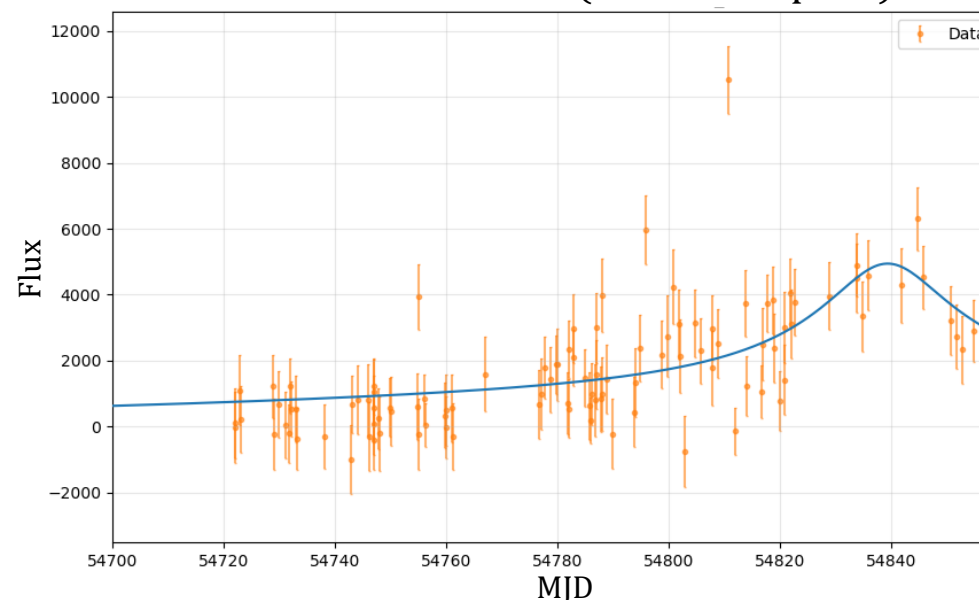
3. Other phenomena (Belokurov et al., 2005)

- Long period variables (e.g., Miras)
- Semiregular variables (SRVs)

Long period & irregularity can mimic microlensing-like light curves.

Inconclusive: Multifrequency data will be helpful since microlensing is colourless.

lc4762_iv: PSPL Fit (zoomed at peak)



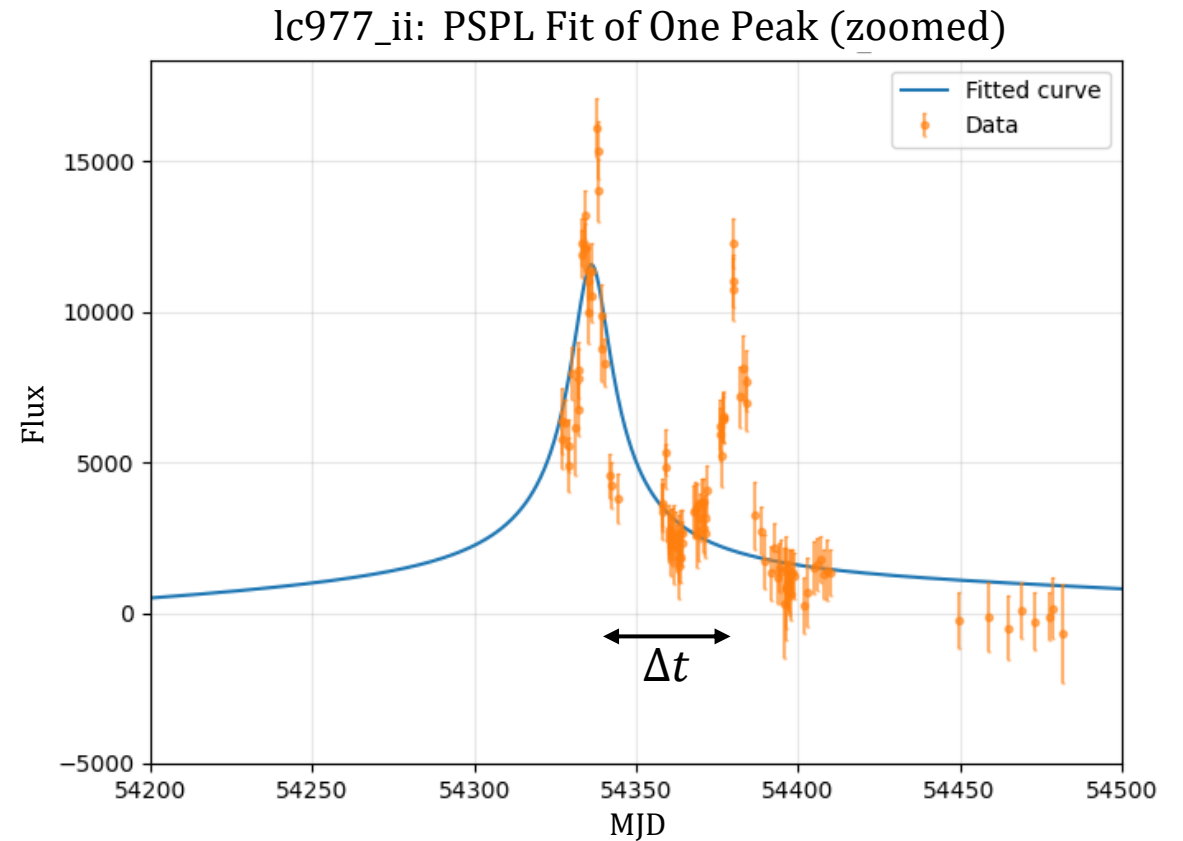
SRVs example from OGLE survey

Part IVb: Non-PSPL Case Studies

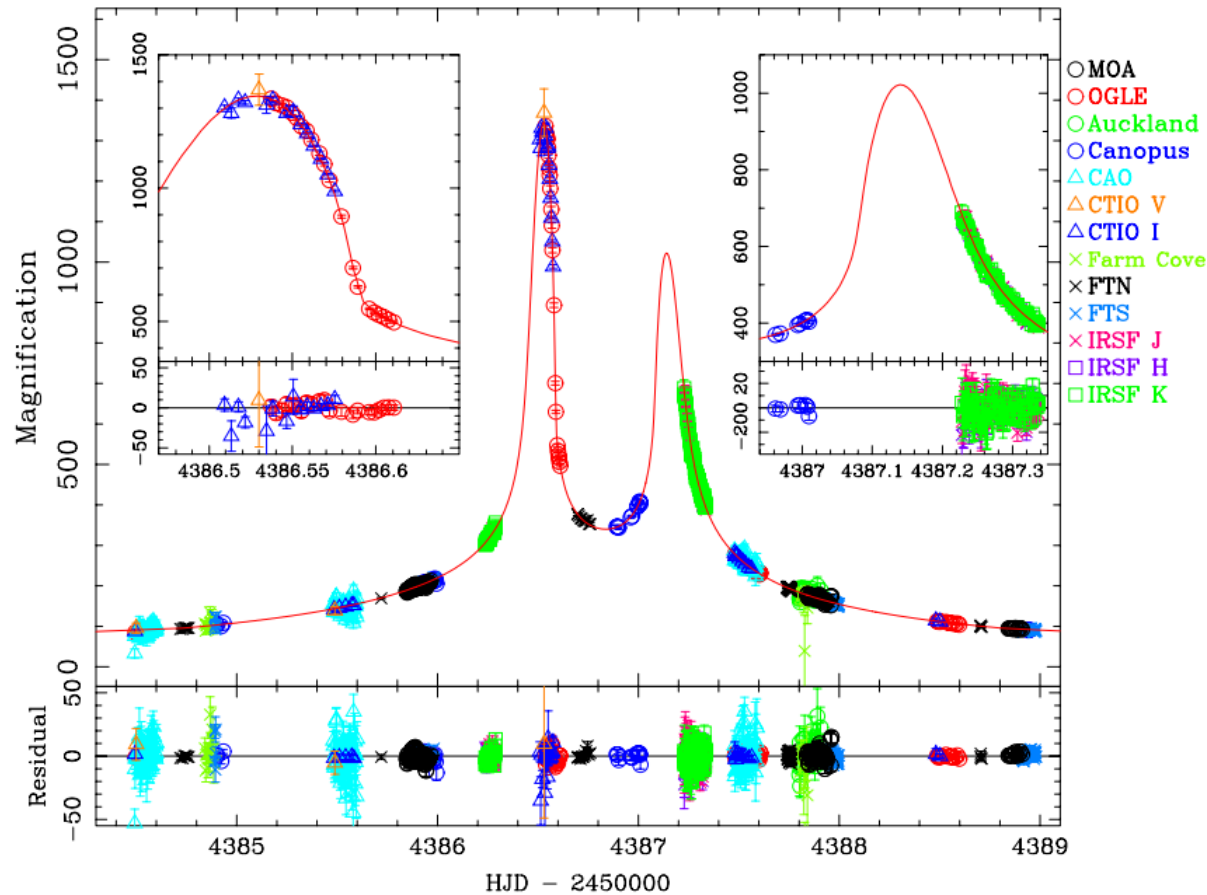
Aim: Analyse light curves for two light curves not resembling the PSPL light curve

Pixel 2: Double Peak

- Each peak is symmetric. Can it be two independent microlensing?
- Fit each peak individually and found the mass of lens.
- Probability of two independent PSPL microlensing events occurring within Δt :
 $\sim 10^{-5} - 10^{-6} \rightarrow$ extremely unlikely
(From Poisson statistics and M31 optical depth)
- More likely a binary-lens system



Pixel 2: Double Peak



Example of binary microlensing with a fitted curve. It looks similar to the light curve we obtained

Magnification of the binary microlensing event, OGLE-2007-BLG-514, Miyake et al (2012).

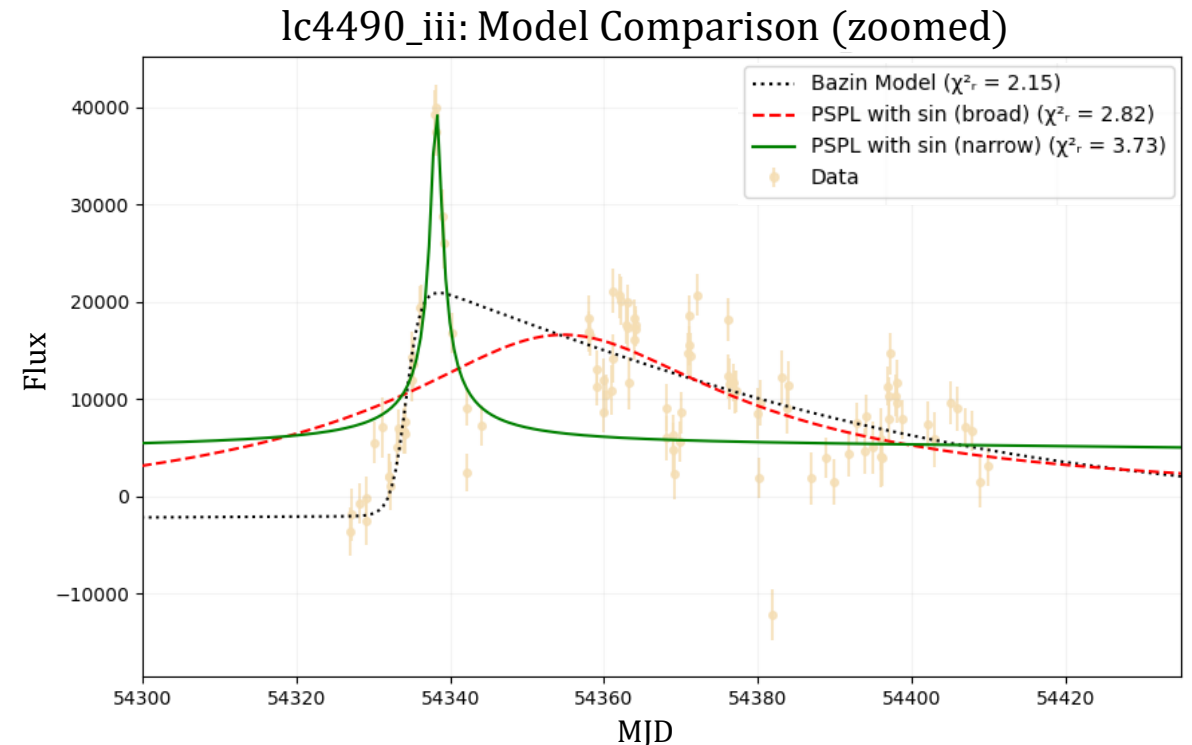
Pixel 3 - Asymmetric Light Curve

- It shows clear asymmetric light curve

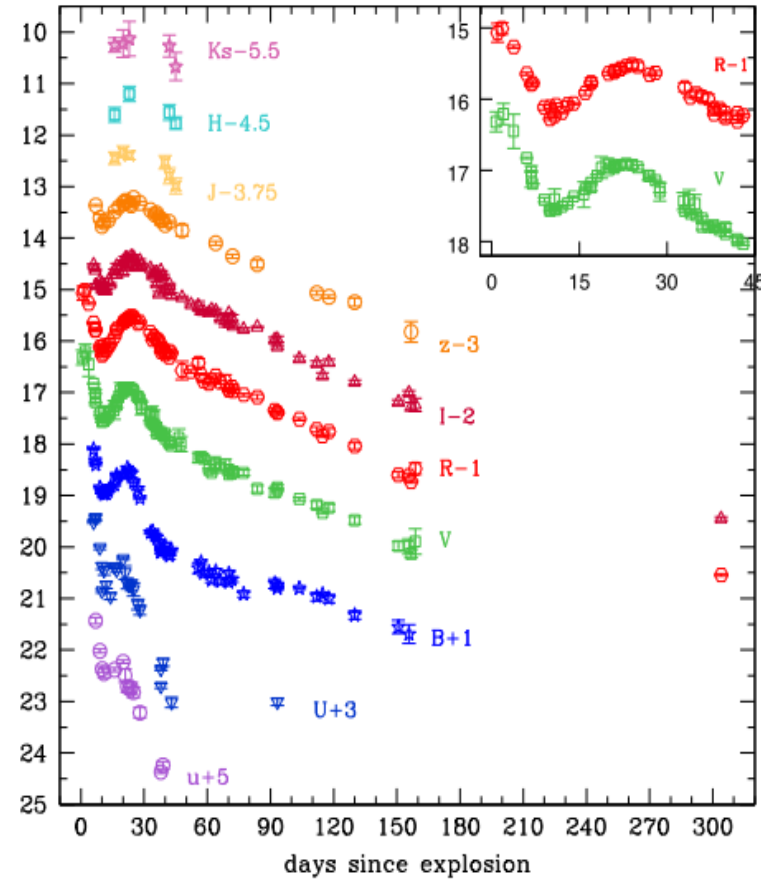
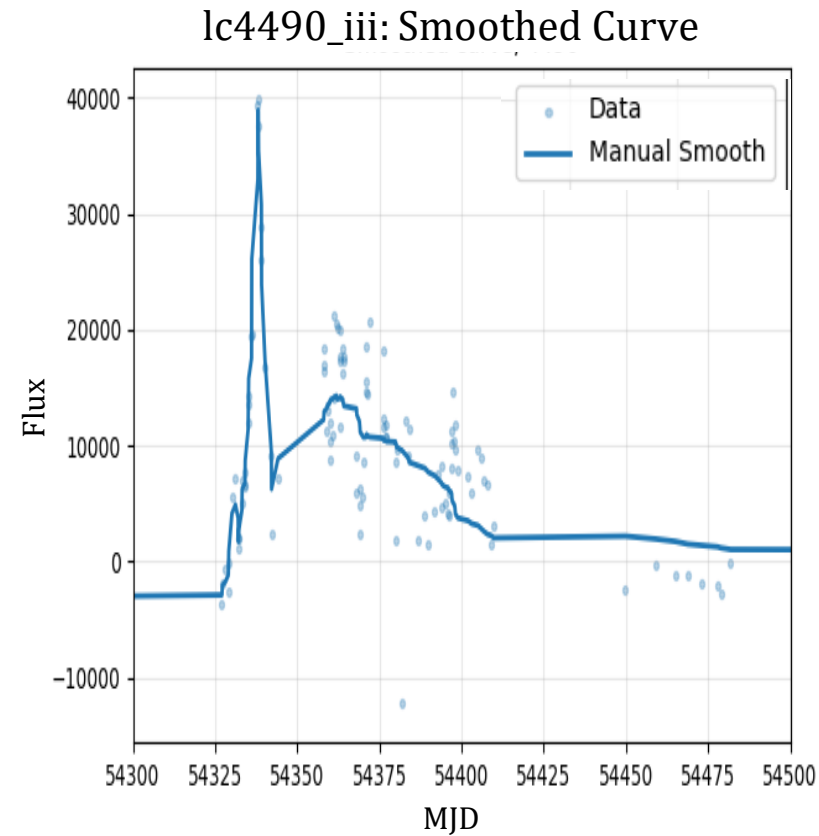
Bazin Model (allows asymmetry):

$$\Delta F(t) = \Delta F_0 \frac{e^{-(t-t_0)/\tau_{\text{fall}}}}{1 + e^{(t-t_0)/\tau_{\text{rise}}}}$$

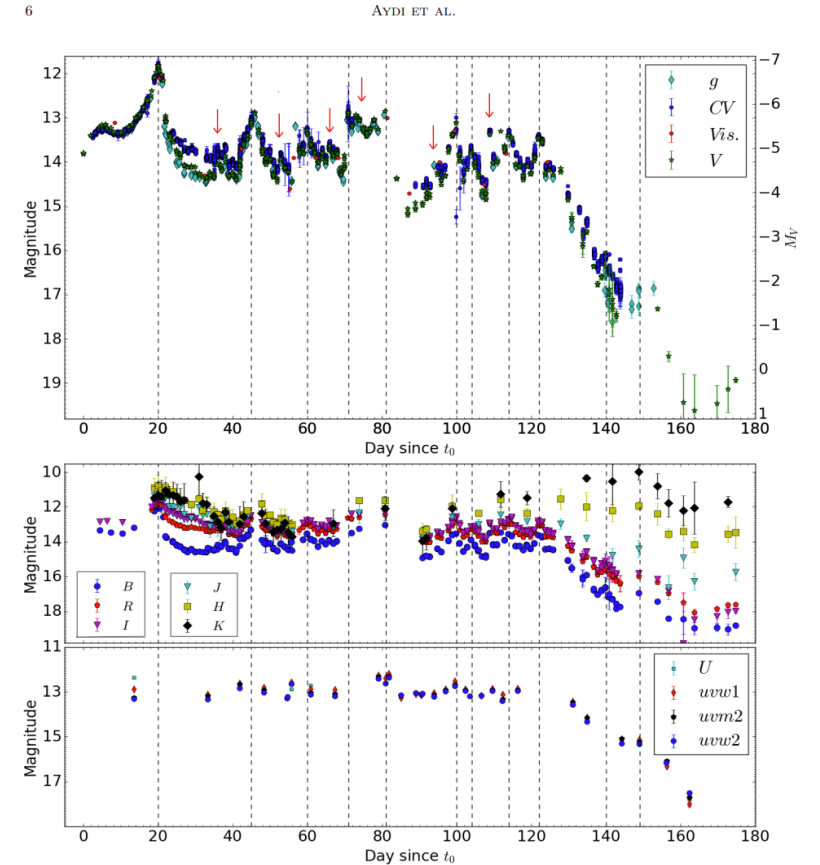
- $\tau_{\text{rise}} = (1.0 \pm 0.1)$ days
- $\tau_{\text{fall}} = (57 \pm 3)$ days
- Bazin Model gives lower χ_r^2 than PSPL
- This is typical behaviour of a transient, with a sharp rise and slower tail.
- However, the data also shows multiple peaks on top of the asymmetry. Is this a nova or supernova?



- Smoothing gives a light curve with double peak.
 - Certain type of novae and supernovae can show multiple peaks
 - It's probably a nova since supernova is extremely rare and much brighter than the galactic bulge
- Note: Different smoothing windows were used for different regions*



Apparent magnitude of Type IIb SN2011fu in multi-frequency range, A. Morales-Garoffolo et al. (2015).



Apparent magnitude of slow nova ASASSN-17pf in multi-frequency range, showing multiple peaks E. Aydi, et al. (2019).

Conclusions

- Built high-quality reference image
- Optimised DIA kernel with dual annealing
- Produced variability map and identified diverse astrophysical events:
 - 2 PSPL microlensing candidates
 - 1 probable binary-lens microlensing
 - 1 nova-like transient
- Substellar and stellar objects can cause microlensing in M31 bulge
- Shows success of DIA for finding low-mass objects in crowded fields
- Observing at multiple wavelengths could enhance analyses

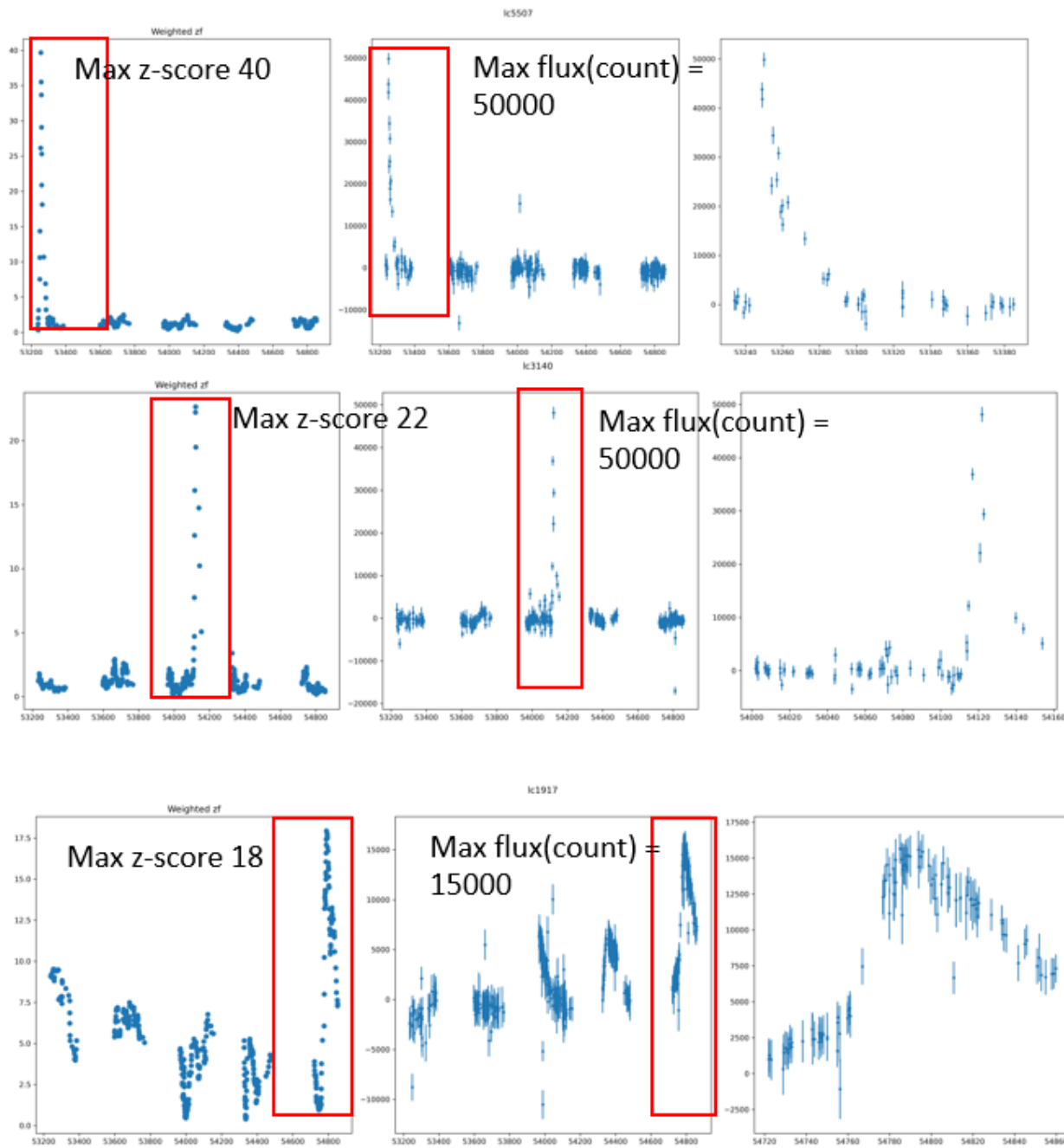
Extension: Identifying candidate light curves

Aim: Find other interesting light curves in the Angstrom data

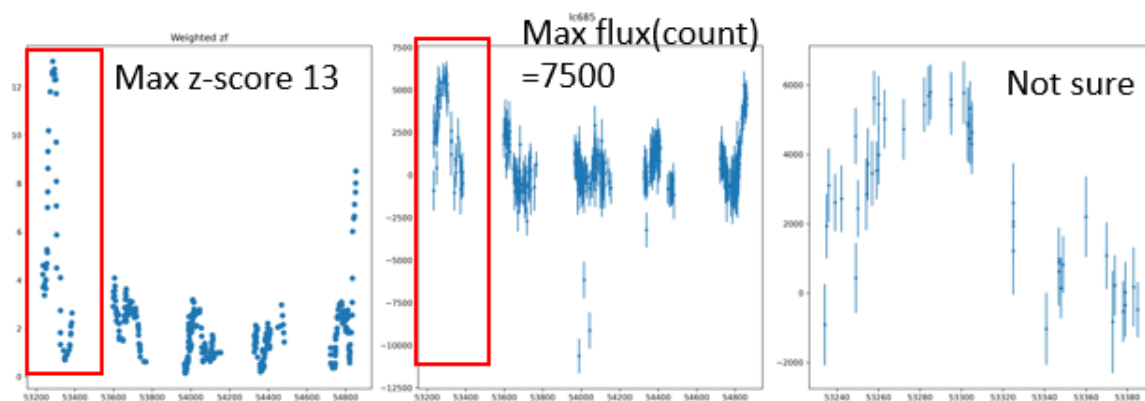
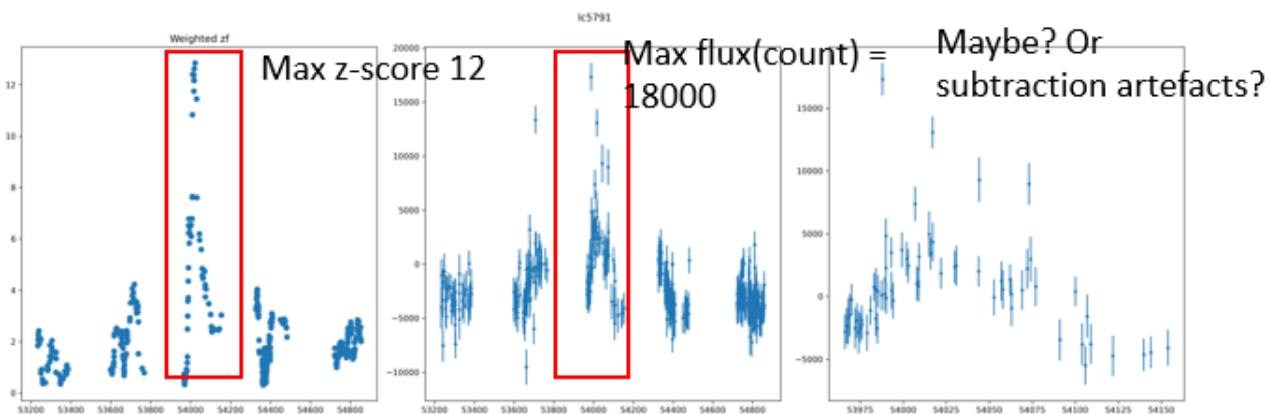
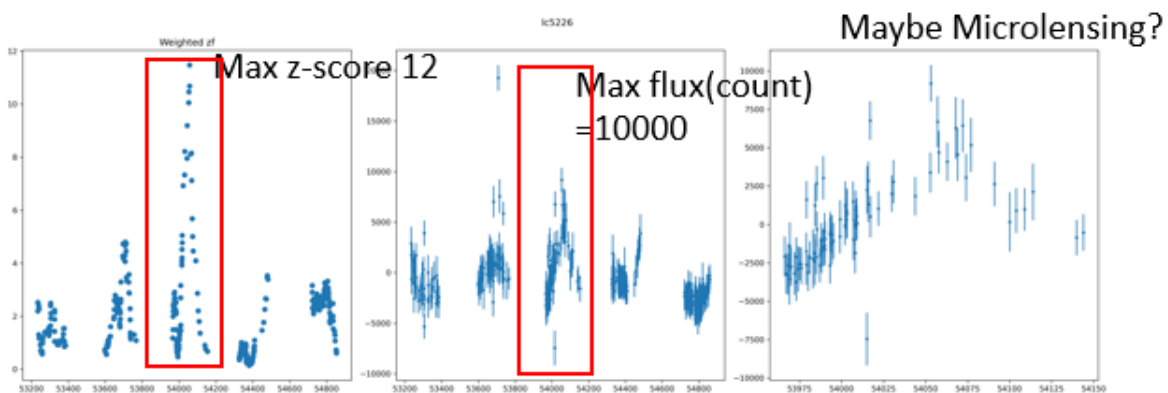
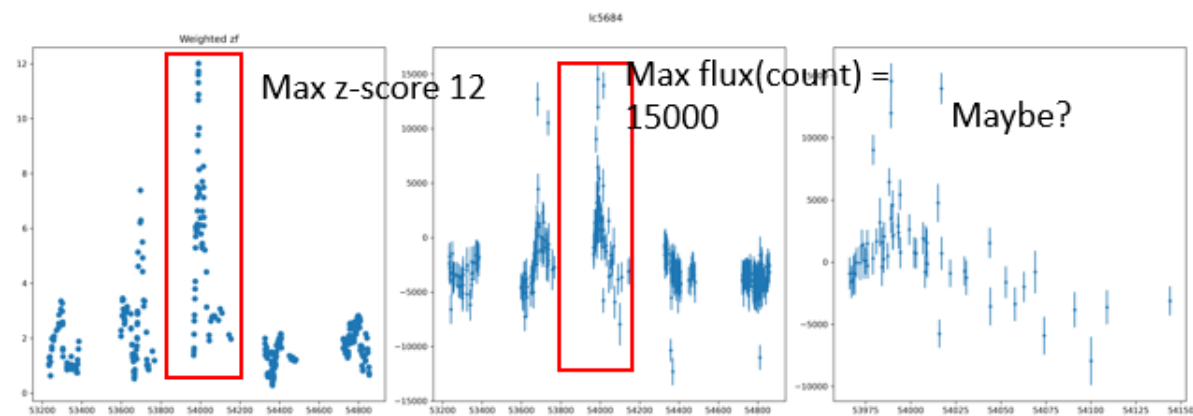
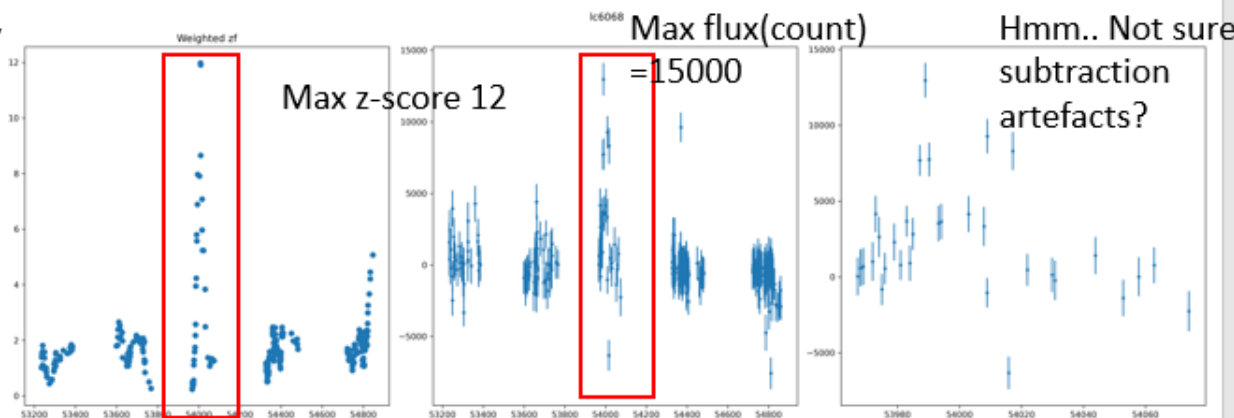
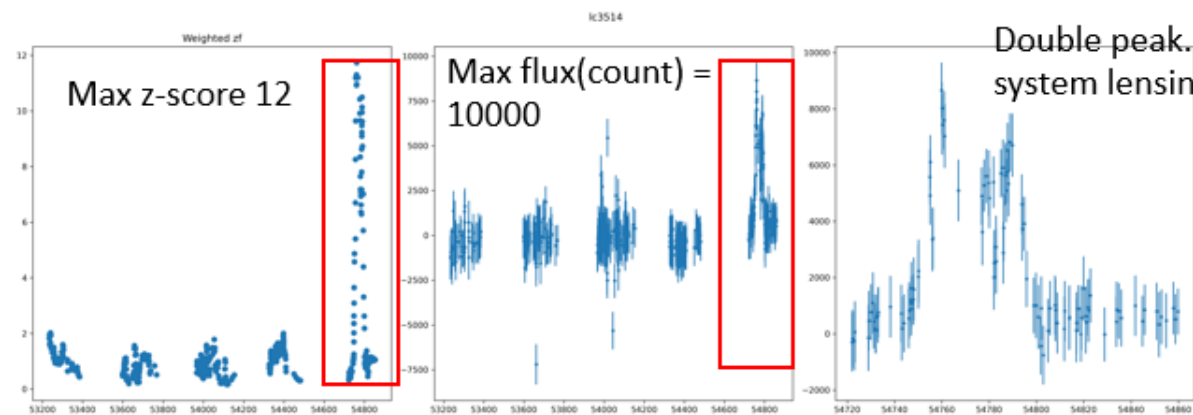
Method to identify peaked light curves

1. Find z-score of each points in light curve
2. Smooth with geometric mean with finite window size. This is to suppress outliers.
3. Rank images using maximum z-score
4. Visually inspect the highest-ranking frames

*Rejected light curves with very large scatters on z-score itself



- These 3 light curves all show sharp, asymmetric curve
- They are all nova-like



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